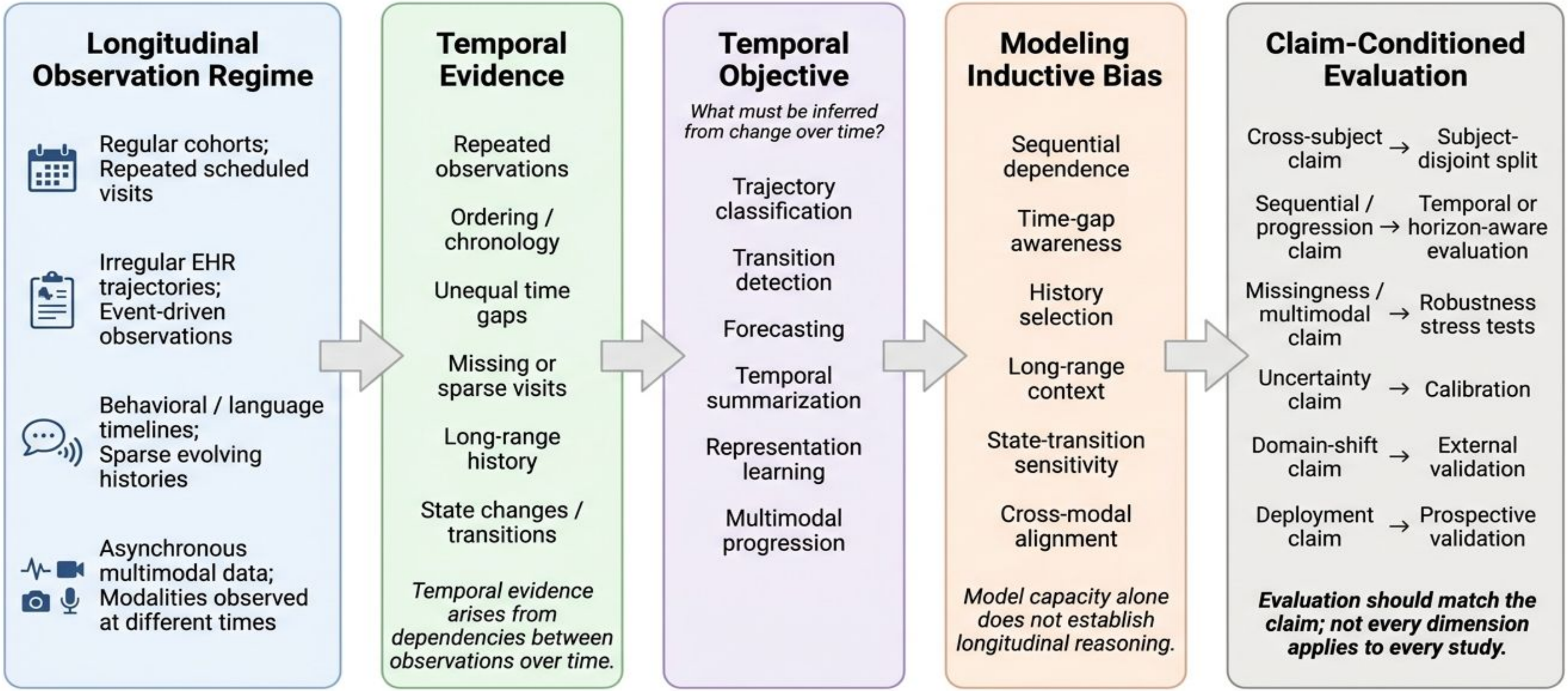
Longitudinal Observation Regime
Regular cohorts; Repeated scheduled visits
Irregular EHR trajectories; Event-driven observations
Behavioral / language timelines; Sparse evolving histories
Asynchronous multimodal data; Modalities observed at different times
Temporal Evidence
Repeated observations
Ordering / chronology
Unequal time gaps
Missing or sparse visits
Long-range history
State changes / transitions
Temporal evidence arises from dependencies between observations over time.
Temporal Objective
What must be inferred from change over time?
Trajectory classification
Transition detection
Forecasting
Temporal summarization
Representation learning
Multimodal progression
Modeling Inductive Bias
Sequential dependence
Time-gap awareness
History selection
Long-range context
State-transition sensitivity
Cross-modal alignment
Model capacity alone does not establish longitudinal reasoning.
Claim-Conditioned Evaluation
Cross-subject claim → Subject-disjoint split
Sequential / progression claim → Temporal or horizon-aware evaluation
Missingness / multimodal claim → Robustness stress tests
Uncertainty claim → Calibration
Domain-shift claim → External validation
Deployment claim → Prospective validation
Evaluation should match the claim; not every dimension applies to every study.